Gradient-Enhanced Gaussian Process Regression for Non-Intrusive POD-Based Reduced-Order Models: An Adjoint-Informed 1D Diffusion Study

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ABSTRACT

Non-intrusive reduced-order models (ROMs) built from proper orthogonal decomposition (POD) and Gaussian process regression (GPR) replace an expensive coupled-physics solve with a cheap parameter-to-quantity-of-interest (QoI) surrogate, but a value-only GP extracts a single scalar per expensive training solve. Gradient-enhanced GPR (GE-GPR) is a well-established remedy in the surrogate-modeling literature; what this paper contributes is the route by which the gradients are obtained. Holding the POD basis fixed turns each retained POD coefficient into a scalar QoI, so a single adjoint solve per retained mode returns the exact derivative of that coefficient with respect to *every* active parameter at once, reusing the factorization of the forward system already paid for. Two consequences follow. Gradient cost scales with the retained rank R, not with the number of parameters p. And, unlike gradient-enhanced POD surrogates that differentiate the snapshots themselves, no derivative of the full snapshot field is ever required, which is what makes the recipe portable to production transport codes where adjoint solves are routine but derivative fields are not available. The construction is verified against exact adjoint gradients on two 1D one-group diffusion test cases, used as a sandbox for scoping the method rather than as physics results in their own right: piecewise-constant material layers and a smooth void/Teflon interface. Combining both into one 5-parameter case at a matched regression budget of M=80, GE-GPR cuts the median field-reconstruction error by 77% relative to standard GPR (6.8×10^{-4} vs. 3.0×10^{-3}) and lands roughly 3.5× above the POD projection floor, versus $\sim$ 15× for standard GPR; repeating over five independent Latin hypercube designs confirms the separation is not an artifact of one design. On the material parameter whose coefficient map is hardest to learn from values alone, GE-GPR reaches standard GPR's converged accuracy with roughly an order of magnitude fewer training solves. A SCEPTRE-based SGEMP charge-deposition problem provides a secondary, real-code check: a 1000-case void-thickness sweep gives a converged standard-GPR baseline (held-out relative error $\varepsilon_q \approx 1.15 \times 10^{-3}$). SCEPTRE's own adjoint output exists for this case set but is intentionally not yet wired into the pipeline, and extending GE-GPR to it is targeted for the December review.

Keywords: reduced-order modeling, proper orthogonal decomposition, Gaussian process regression, adjoint sensitivity, gradient-enhanced surrogate

1. INTRODUCTION

Coupled-physics solves are often too expensive to call repeatedly inside a design loop, an uncertainty-quantification study, or an optimization routine. Non-intrusive reduced-order models (ROMs) address this by replacing the expensive solver with a cheap surrogate map from input parameters μ to a quantity of

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interest (QoI). A common recipe is proper orthogonal decomposition (POD): a snapshot matrix of the QoI, evaluated at a set of training parameter values, is factored by a thin singular value decomposition (SVD) to give a low-rank basis Ψ, the QoI at any new μ is approximated as $y(\mu) \approx \Psi\alpha(\mu)$, and a Gaussian process (GP) is fit to learn the map $\mu \mapsto \alpha(\mu)$ for each retained mode.

The limitation of this recipe is that a value-only GP is trained on function values alone, so each expensive training solve contributes a single scalar per POD mode. Supplying derivative observations alongside the values is a well-established remedy. Gaussian process regression on derivative data dates to Morris, Mitchell and Ylvisaker [1], and gradient-enhanced kriging (GEK) or gradient-enhanced GPR (GE-GPR) has since been developed extensively in the design-optimization and surrogate-modeling literature [2, 3, 4, 5, 6, 7]. Zimmermann [8] in particular combines gradient enhancement with POD. The GE-GPR estimator itself is therefore not claimed as new here.

What this paper contributes is the route to the gradients, and what that route costs. The POD basis Ψ is computed once and held fixed, which turns each retained coefficient $\alpha_k(\mu) = \psi_k^\top H\phi(\mu)$ into an ordinary scalar QoI, linear in the state. Classical adjoint sensitivity analysis [9] then applies directly: one adjoint solve per retained mode returns $\partial\alpha_k/\partial\mu_j$ for *every* active parameter μ_j simultaneously, reusing the factorization of the forward system already computed for $\phi(\mu)$. Three consequences distinguish this from the GEK literature above. First, *gradient cost scales with rank, not dimension*: a training sample costs R adjoint solves, where R is the retained POD rank (here 1–2), and yields $R \times p$ derivative observations, so adding a parameter adds no solves. Finite-difference or tangent-mode gradient enhancement instead costs $O(p)$ extra solves per sample, which is what limits GEK in high dimension. Second, *no snapshot derivative fields are needed*: gradient-enhanced POD surrogates in the vein of Ref. [8] require the derivative of the snapshot field itself, $\partial y/\partial\mu \in \mathbb{R}^{N_y}$, which a production transport code generally does not expose, whereas the adjoint delivers the R coefficient derivatives directly and never forms $\partial y/\partial\mu$. This matters in the nuclear-engineering setting the method is aimed at, where adjoint solves are routine machinery (shielding, detector response, perturbation theory) but full derivative fields are not. Third, the gradients are *exact*: because α_k is linear in ϕ, the sensitivity is algebraically exact to solver tolerance, so any measured sample-efficiency gain reflects the information content of the gradients rather than finite-difference step-size tuning.

A one-group, 1D diffusion solver serves as a sandbox for scoping this recipe rather than as the paper's object of interest in its own right. Because its adjoint sensitivities can be verified in closed form and by finite differences, every sample-efficiency claim can be checked exactly before the recipe is applied to a harder code, here SCEPTRE, an export-controlled Sandia radiation-transport code (Sec. 5.5), where it cannot be checked the same way. SCEPTRE does produce adjoint sources for this case set, but integrating them into GE-GPR is left for future work. The pipeline (forward solve → POD → adjoint → GE-GPR) is not specific to the diffusion solver; extending it to the full 3D coupled-physics stack is targeted for the December review, time permitting.

Concretely, this paper derives the adjoint sensitivities of the POD coefficients for the 1D diffusion solver, both for piecewise-constant material layers and for a smooth, differentiable moving-interface geometry; measures the sample efficiency of GE-GPR against standard, value-only GPR on both test cases at matched regression budget and fixed POD basis, and over repeated independent Latin hypercube designs; combines the two cases' parameters into one 5-parameter study to show how the relative benefit changes as the number of active parameters grows; and establishes a standard-GPR accuracy baseline on a real, export-controlled SCEPTRE dataset as the first step toward extending GE-GPR to it.

2. 1D DIFFUSION SANDBOX: FORWARD MODEL AND BASELINE ROM

The forward model and baseline ROM are the gradient-free scaffolding GE-GPR is verified against, and are kept brief. The solver is a 1D, one-group neutron diffusion problem,

$$-\frac{d}{dx}\left[D(x)\frac{d\phi}{dx}\right] + \Sigma_a(x)\,\phi(x) = q(x), \qquad x \in (0, L), \tag{1}$$

with Dirichlet boundary conditions, discretized with a cell-centered, second-order finite-volume scheme on a uniform N-cell mesh into a sparse linear system $A(\mu)\phi(\mu) = s(\mu)$. Two parameterizations of μ are used, corresponding to the two test cases carried through the rest of the paper:

- **Material layers:** μ is the per-layer diffusion coefficient, absorption cross section, and source, $(D_m, \Sigma_{a,m}, q_m)$, piecewise constant over N_{mat} layers.
- **Void/Teflon geometry:** a symmetric slab (Shield | Teflon | Void | Core | Core | Void | Teflon | Shield) in which the Void/Teflon interface is the only feature allowed to move. Because a hard material boundary is not differentiable in its own location (moving it changes which mesh cell it falls in, not a smooth function of a coordinate), the hard step is replaced, at that interface only, by a logistic switch $\sigma(x; a, b) = \left[1 + e^{-a(x-b)}\right]^{-1}$, where b is the interface location ($\sigma(b) = 0.5$ for any a) and a is the transition steepness. The Core/Void and Teflon/Shield boundaries are left as hard steps, since they do not move in this study.

How D, Σ_a and q are interpolated. All three coefficients are stored as cell-wise arrays evaluated at cell centers x_i; no sub-cell averaging is performed. In the material-layer case they are piecewise constant, each cell taking its layer's value, so a layer parameter enters every cell it owns identically. In the geometry case each of the three is blended across the moving interface by the same switch,

$$v(x_i) = v_{\text{void}} + (v_{\text{Teflon}} - v_{\text{void}})\,\sigma(r_i; a, b)\,, \qquad r_i = |x_i - L/2|, \tag{2}$$

for $v \in \{D, \Sigma_a, q\}$ within the shell $r_i \in [r_{\text{core}}, r_{\text{core}} + w_{\text{shell}}]$, and the fixed core and shield values outside it. Using one switch for all three keeps a single (a, b) pair, so the same two derivatives $\partial\sigma/\partial a$ and $\partial\sigma/\partial b$ propagate to all three coefficient fields. In the assembled system Σ_a and q enter cell-locally, while D enters only through the interior-face conductances $\beta_{i+1/2} = 2d_i d_{i+1}/(d_i + d_{i+1})$ with $d_i = D_i/\Delta x_i$, a harmonic mean of half-cell conductances.

A masked flux window $y(\mu) = H\phi(\mu)$ over a fixed sub-interval of cells is the QoI. The snapshot matrix $Y \in \mathbb{R}^{N_y \times M_{\text{POD}}}$ is factored by a thin SVD with the retained rank R set by an energy cutoff η on the singular-value spectrum, giving the POD basis $\Psi \in \mathbb{R}^{N_y \times R}$ and coefficients $\alpha(\mu) = \Psi^\top y(\mu)$. The baseline surrogate is one independent ARD-RBF-kernel GP per retained mode (scikit-learn's `GaussianProcessRegressor`) trained on $\{\mu^{(i)}, \alpha_k(\mu^{(i)})\}$. The field is reconstructed as $\hat{y}(\mu) = \Psi\hat{\alpha}(\mu)$, and every result is reported as the relative field-reconstruction error

$$\varepsilon_y(\mu) = \frac{\|\hat{y}(\mu) - y(\mu)\|_2}{\|y(\mu)\|_2} \tag{3}$$

on a held-out test set. The reference point is the POD projection-error floor $\varepsilon_{\text{proj}}(\mu) = \|\Psi\Psi^\top y(\mu) - y(\mu)\|_2 / \|y(\mu)\|_2$, the best any surrogate built on this basis could do, since it is the error from truncating to rank R alone, independent of how well the GP fits $\alpha(\mu)$.

3. ADJOINT SENSITIVITIES OF THE POD COEFFICIENTS

Each POD coefficient is a scalar QoI that is linear in the state, $\alpha_k = \psi_k^\top y = \psi_k^\top H\phi$, so the classical adjoint construction [9] applies without modification. Differentiating the residual $R(\phi, \mu) = A(\mu)\phi - s(\mu) = 0$ with

respect to μ_j and eliminating $\partial\phi/\partial\mu_j$ with an adjoint variable gives the central relation of this paper,

$$\boxed{\frac{\partial \alpha_k}{\partial \mu_j} = \lambda_k^\top \left(\frac{\partial s}{\partial \mu_j} - \frac{\partial A}{\partial \mu_j}\,\phi \right), \qquad A^\top \lambda_k = H^\top \psi_k.} \tag{4}$$

The left-hand system is solved once per retained mode k, using the same factorization of A (transposed) already computed for the forward solve $A\phi = s$. The right-hand contraction is then a matrix-vector product per parameter, so all p partials of mode k follow from that one solve. A training sample therefore costs R adjoint solves and yields $R \times p$ derivative observations, independent of p. The same λ_k also reproduces the coefficient itself, since $\alpha_k = (A^\top \lambda_k)^\top \phi = \lambda_k^\top s$, which is used below as a verification identity. Everything case-specific then reduces to supplying $\partial A/\partial\mu_j$ and $\partial s/\partial\mu_j$.

Material-layer case. For the per-layer source and absorption, Eq. (4) collapses to closed forms, since $\partial s/\partial q_m$ and $\partial A/\partial \Sigma_{a,m}$ are cell-local and supported only on layer m. With Δx_i the cell width, layer(i) the layer index of cell i, and interior cells excluded where a Dirichlet row overwrites the residual:

$$\frac{\partial \alpha_k}{\partial q_m} = \sum_{i \,\in\, \text{layer}\, m,\ \text{interior}} (\lambda_k)_i \, \Delta x_i, \tag{5}$$

$$\frac{\partial \alpha_k}{\partial \Sigma_{a,m}} = - \sum_{i \,\in\, \text{layer}\, m,\ \text{interior}} (\lambda_k)_i \, \phi_i \, \Delta x_i. \tag{6}$$

The diffusion-coefficient sensitivity requires more work, since D enters only through the face conductances. With $\partial\beta_{i+1/2}/\partial D_i = 2d_{i+1}^2/[(d_i + d_{i+1})^2 \Delta x_i]$ (and symmetrically for D_{i+1}), $\partial\alpha_k/\partial D_m = -\lambda_k^\top(\partial A/\partial D_m)\phi$ summed over the faces bordering layer m.

Geometry case. Differentiating the blend of Eq. (2) through the logistic switch gives smooth per-cell "direction vectors" $\partial(D, \Sigma_a, q)/\partial a$ and $\partial(D, \Sigma_a, q)/\partial b$ in place of the layer indicator used above; assembling $\partial A/\partial\mu_j$ and $\partial s/\partial\mu_j$ from them, Eq. (4) applies unchanged. Because the direction-vector construction is generic to any per-cell derivative field, the same code that handles the geometry case's two parameters (interface location b, steepness a) extends without modification to additional material parameters promoted from fixed constants to active dimensions (Sec. 5, combined case).

Verification. Every sensitivity above is checked three ways, identically in both test cases: an *identity check* comparing $\alpha_k = \lambda_k^\top s$ against direct projection $\alpha_k = \psi_k^\top y$, which must agree to solver tolerance; a *finite-difference check* against a centered difference over a small parameter perturbation; and a *1D sweep* of $\alpha_k(\mu)$ with the adjoint-predicted tangent line overlaid at several expansion points. The identity and finite-difference checks run on every case reported below (identity error $\sim 10^{-11}$, finite-difference agreement $\sim 10^{-5}$ relative); the sweep is a per-parameter diagnostic run for the material and geometry cases.

4. GRADIENT-ENHANCED GAUSSIAN PROCESS REGRESSION

This section fixes notation and states the implementation choices the results depend on; the estimator itself is standard [1, 3]. A GP is closed under differentiation: if $\alpha_k(\mu) \sim \mathcal{GP}(0, k(\mu, \mu'))$, the joint distribution of α_k and $\partial\alpha_k/\partial\mu$ at any finite training set is itself Gaussian, with covariance blocks obtained in closed form by differentiating the kernel. Training on values and gradients together is therefore not an approximation added to a value-only GP; it is the same model, conditioned on more of what it already implies.

Feeding gradients in as extra scalar features, or fitting a separate model per derivative channel, would break that single joint covariance: those channels would be treated as independent, discarding the cross-covariance $\mathrm{Cov}(\alpha_k, \partial\alpha_k/\partial\mu_m)$ that makes a gradient observation informative about *nearby* function values rather than

only the training point itself. Stacking $y_{\text{aug}} = [\alpha_k;\ \partial\alpha_k/\partial\mu]$ under one augmented kernel is what preserves it.

Concretely, for an ARD-RBF kernel $k(x,x') = \sigma_f^2 \exp\!\left(-\frac{1}{2}\sum_j r_j^2/\ell_j^2\right)$ with $r_j = x_j - x'_j$, the three covariance blocks needed are

$$\operatorname{Cov}(f(x), f(x')) = k(x,x'), \tag{7}$$

$$\operatorname{Cov}\left(f(x), \frac{\partial f}{\partial x'_m}(x')\right) = k(x,x') \cdot \frac{r_m}{\ell_m^2}, \tag{8}$$

$$\operatorname{Cov}\left(\frac{\partial f}{\partial x_m}(x), \frac{\partial f}{\partial x'_n}(x')\right) = k(x,x') \cdot \left[\frac{\delta_{mn}}{\ell_m^2} - \frac{r_m r_n}{\ell_m^2 \ell_n^2}\right], \tag{9}$$

assembled into the block Gram matrix

$$K_{\text{aug}} = \begin{bmatrix} K_{ff} + \sigma_{nf}^2 I & K_{fd} \\ K_{fd}^\top & K_{dd} + \sigma_{nd}^2 I \end{bmatrix}, \tag{10}$$

of size $M(1+p_{\text{act}})$ for M training points and p_{act} active gradient dimensions, fit by maximizing the augmented marginal likelihood. The same hyperparameters $(\sigma_f, \ell_1, \ldots, \ell_{p_{\text{act}}})$ now have to explain both the values and their slopes simultaneously, which is what lets a handful of gradient observations pin down length scales that would otherwise need many more function evaluations. These identities are implemented directly in `sklearn_implementation/ge_gpr.py`.

The cost is a Cholesky factorization of $O\big((M(1+p_{\text{act}}))^3\big)$, and the augmented matrix can become ill-conditioned when length scales shrink relative to the training-point spacing, since the gradient-gradient block scales as $1/\ell^2$; both bound how far the method scales in p_{act} (Sec. 6).

Scaling, and the chain rule it forces. Inputs are min–max normalized to $[0,1]^p$ and each mode's coefficients are standardized to zero mean and unit variance over the training set, $z_k = (\alpha_k - \bar{\alpha}_k)/\sigma_{\alpha_k}$. The adjoint gradients are computed in physical units and must be transformed consistently into that space before they are used as observations,

$$\frac{\partial z_k}{\partial x_j} = \frac{\partial \alpha_k}{\partial \mu_j} \cdot \frac{\mu_j^{\max} - \mu_j^{\min}}{\sigma_{\alpha_k}}, \tag{11}$$

with $x_j = (\mu_j - \mu_j^{\min})/(\mu_j^{\max} - \mu_j^{\min})$. This is worth stating explicitly because it is a silent failure mode: an inconsistently scaled gradient is still a valid-looking number, and a GE-GP trained on one will fit, report a finite marginal likelihood, and simply predict badly. A value-only GP has no such coupling and so gives no warning.

Hyperparameters and conditioning. The quantities σ_{nf} and σ_{nd} in K_{aug} are independent observation-noise standard deviations, on the standardized function values and on the scaled derivative observations of Eq. (11) respectively. They are not fixed: both are learned alongside $(\sigma_f, \ell_1, \ldots, \ell_{p_{\text{act}}})$ by maximizing the augmented log marginal likelihood with L-BFGS-B in log space, initialized at 10^{-3} and bounded to $[e^{-10}, 1] \approx [4.5 \times 10^{-5}, 1]$, with $\log\sigma_f$ and $\log\ell_j$ bounded to $[-5, 5]$. Keeping them separate lets the fit express that values and slopes carry different effective noise once scaled. Distinct from these, a *fixed* jitter of 10^{-6} is added to the diagonal of K_{aug} purely for Cholesky conditioning, and is not optimized. The value-only baseline uses the matching scikit-learn kernel (constant × ARD-RBF + white noise, noise variance bounded below at 10^{-6}) on the same standardized targets, so that any difference in accuracy isolates the effect of adding gradients.

The implementation (`GEGaussianProcess`) is our own, since scikit-learn's `GaussianProcessRegressor` cannot take derivative observations; this is an enabling detail rather than a contribution. The same class is

reused unmodified across every diffusion case below, and does not care what μ physically means or where the gradients came from, only that they are $\partial\alpha_k/\partial\mu$ expressed in the scaled space of Eq. (11).

5. RESULTS

What is held fixed, and what the training budget means. Every case below uses a uniform $N = 200$-cell mesh, a QoI window $x \in [3, 7]$, an energy-cutoff threshold $\eta = 99.999\%$ for the POD rank, and Latin hypercube (LHS) sampling of μ, so results are on a directly comparable footing. Two sample counts must be kept distinct. M_{POD} snapshots build the POD basis Ψ once, by thin SVD, and that basis is then frozen; M of those same snapshots (the first M of the LHS design, so that increasing M nests) fit the surrogate, with GE-GPR and standard GPR seeing exactly the same M points. A third, disjoint LHS set of M_{test} snapshots is drawn for held-out evaluation and never enters either the basis or the fit. The remaining $M_{\mathrm{POD}} - M$ training snapshots are *not* a test set; they contribute to the basis only. Every learning curve below is therefore a **fixed-basis regression sample-efficiency study**: it measures how many solves the regression stage needs on a basis that is already converged, and deliberately does not charge basis construction to either surrogate. Total ROM cost, where basis and regression share one shrinking budget, is a different and harder question left to future work. The projection floor $\varepsilon_{\mathrm{proj}}$ quoted alongside every result is the accuracy that basis permits, and is the correct reference against which a regression gain should be read. Table I gives the per-case budgets together with the results; the combined case's smaller M is a cost decision, since at $p_{\mathrm{act}} = 5$ the augmented Gram matrix is already 6× wider than the value-only one at equal M.

5.1. Material-Layer Case

Each of the three per-layer parameters ($D \in [0.5, 2.0]$, $\Sigma_a \in [0.05, 1.5]$, $q \in [0.0, 2.0]$, all others fixed) was run as an independent single-parameter study, so $\alpha_k(\mu)$ is a genuine 1D curve and the only question is whether the adjoint gradient gives GE-GPR an edge fitting it. Table I collects the converged result for this and every other diffusion case.

Table I. Sample budgets and median held-out field-reconstruction error for every diffusion case. Each case's POD projection floor $\varepsilon_{\mathrm{proj}}$ is given in parentheses after its parameter name. M is the matched regression budget at which the errors are quoted (the largest run for that case) and R the POD rank retained at $\eta = 99.999\%$. Rows marked † are at their floor for both surrogates, so their improvement entry is uninformative.

Active parameter(s)		p_{act}	M_{POD}	M_{test}	M	R	Std GPR	GE-GPR	improvement
Material layers, one at a time									
D	$(\varepsilon_{\mathrm{proj}} = 3.03 \times 10^{-5})^{\dagger}$	1	500	100	250	2	3.06×10^{-5}	3.03×10^{-5}	1.0%
Σ_a	(1.05×10^{-4})	1	500	100	250	2	1.58×10^{-4}	1.05×10^{-4}	33.6%
q	(2.72×10^{-7})	1	500	100	250	1	3.00×10^{-5}	1.38×10^{-6}	95.4%
Void/Teflon geometry									
void_thickness	$(8.84 \times 10^{-4})^{\dagger}$	1	300	60	250	2	8.84×10^{-4}	8.84×10^{-4}	0.0%
steepness	$(5.30 \times 10^{-4})^{\dagger}$	1	300	60	250	1	5.30×10^{-4}	5.30×10^{-4}	0.0%
both	$(9.32 \times 10^{-4})^{\dagger}$	2	300	60	100	2	9.33×10^{-4}	9.32×10^{-4}	0.1%
Combined									
all five	(1.96×10^{-4})	5	300	60	80	2	2.95×10^{-3}	6.83×10^{-4}	76.8%

The strongest improvement is for the source term q: gradient information yields a 95.4% reduction in median reconstruction error, and the learning curve shows where it comes from. GE-GPR reaches $\varepsilon_y \approx 4.8 \times 10^{-6}$

at M=10, already better than standard GPR's converged M=250 value of 3.0×10^{-5}, so GE-GPR matches or beats standard GPR's accuracy with roughly an order of magnitude fewer training solves. Standard GPR remains above 3×10^{-5} across the whole budget, well short of that case's 2.7×10^{-7} projection floor; GE-GPR closes most of the remaining distance to it.

The $\varepsilon_{\text{proj}}$ column also identifies the apparent $\sim 3 \times 10^{-5}$ floor in the D row: it is the POD projection floor of that case's rank-2 basis, not a property of either regressor, and not the noise kernel (whose optimized level sits well below it), the standardization, or a hyperparameter bound. Both surrogates are at that floor, so the 1.0% "improvement" measures the residual gap standard GPR still has to it, not what gradients are worth; the same applies to Σ_a, where GE-GPR sits exactly at its floor and standard GPR 50% above it. The converged number therefore understates the D case, where gradients help early but not asymptotically: GE-GPR is at the floor from M=10 onward, while standard GPR needs $M \approx 200$ to reach it (5.4×10^{-5} at M=10, 3.4×10^{-5} at M=50). That $\partial \alpha_k / \partial D_m$ is the least informative of the three is consistent with Sec. 3: it is the one sensitivity that is not a simple linear functional of λ_k.

5.2. Geometry Case

With only the interface location b (void_thickness $\in [0.001, 0.25]$) active and steepness fixed at a=200, mode 1 carries 99.99% of the SVD energy and GE-GPR shows essentially *no* benefit: the map is smooth and the design dense enough that a value-only GP already resolves it to the projection floor. Steepness alone gives the same null result, and with both active the two surrogates converge to the same median error at matched M=100. All three geometry rows of Table I sit at their projection floor, so the zeros are saturation rather than failure: on this basis no surrogate could do better, and the comparison has no headroom left to resolve.

The two-parameter row nonetheless hides faster early convergence. At M=10 GE-GPR is at $\varepsilon_y = 1.32 \times 10^{-3}$ against standard GPR's 1.37×10^{-3}, both noisy below $M \approx 20$; GE-GPR settles at the floor by $M \approx 30$–40, while standard GPR does not close the gap until $M \approx 75$–100. Gradients here yield *faster convergence to the POD floor*, not a lower floor, a distinct mechanism from the combined case below, where GE-GPR's advantage persists at every budget.

5.3. Combined Case

All 5 parameters (void_thickness, steepness, and the Teflon shell's D, Σ_a and q) are active simultaneously in one shared forward/adjoint/GE-GPR pipeline, where one adjoint solve per mode still gives all 5 partials at once. At matched M=80, GE-GPR reaches a median $\varepsilon_y = 6.83 \times 10^{-4}$ against standard GPR's 2.95×10^{-3}, a 76.8% reduction. GE-GPR also lands much closer to the projection floor ($\varepsilon_{\text{proj}} = 1.96 \times 10^{-4}$), at $\sim 3.5\times$ the floor versus standard GPR's $\sim 15\times$.

That 76.8% is not the largest relative gain in this paper: the single-parameter q case reaches 95.4% (Table I). The two do not measure the same thing. Each one-at-a-time percentage is capped by how much headroom that case has above its own projection floor, and three of the five have none at all, so their 0% entries say nothing about what gradients are worth. What is distinctive about the combined case is that both surrogates remain well above the floor, so the comparison is resolvable at every budget and GE-GPR's advantage is a persistent error gap rather than a head start that closes.

Figure 1 shows the mechanism. GE-GPR matches standard GPR's converged M=80 result (2.95×10^{-3}) at $M \approx 30$ (1.61×10^{-3}), a factor ~ 2.7 fewer training solves, and the two curves separate from the smallest budget and stay separated. One adjoint solve per mode still yields all 5 partials, while a value-only GP needs proportionally more samples to resolve the higher-dimensional map. That solve reuses the factorization of the same system matrix that produced $\phi(\mu)$, differing only in the right-hand side and a transpose, so at

$p_{\text{act}} = 5$ and $R = 2$ two extra solves per sample deliver ten derivative observations. The price is paid entirely at fit time, in the $M(1 + p_{\text{act}})$-sized augmented kernel.

5.4. Repeatability Across Independent Designs

A learning curve built from nested prefixes of a single LHS design confounds the method with the design: a favorable early prefix can advantage one surrogate. The combined case was therefore repeated over five independent LHS designs (a fresh train/test pair per design, the identical pipeline and budgets otherwise), Fig. 1 (right). At every budget the median improvement across designs stays between 83% and 88%; at M=80, GE-GPR's median ε_y is 6.30×10^{-4} (range 4.92–6.83×10^{-4}) against standard GPR's 4.17×10^{-3} (range 2.30–5.70×10^{-3}), a median improvement of 85.1% (range 72.6–89.0%). Standard GPR's design-to-design spread is roughly 2.5× wider than GE-GPR's, so the single-design result above (76.8%) sits within this range rather than being a favorable outlier.

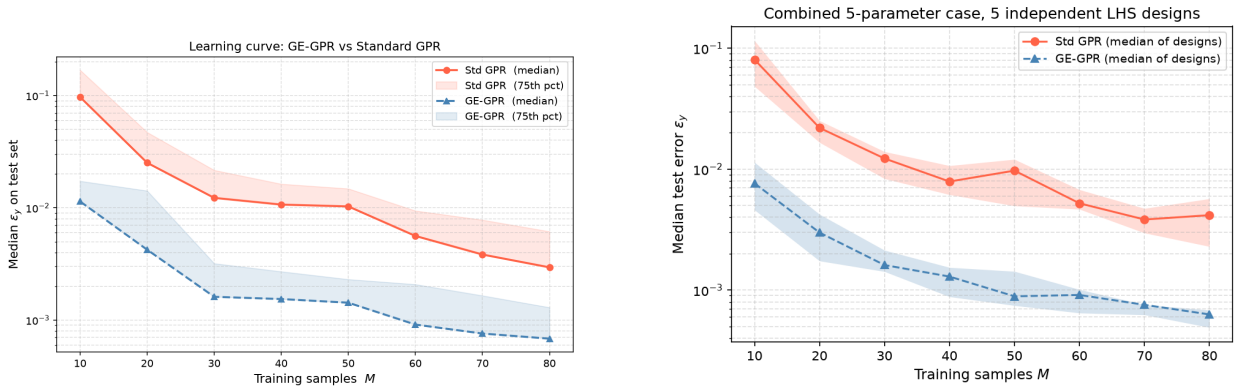


Figure 1. Combined 5-parameter learning curves. *Left*: a single LHS design; GE-GPR reaches standard GPR's M=80 accuracy by $M \approx 30$ and the gap persists rather than closing, with bands giving the 75th-percentile error. *Right*: five independent LHS designs, lines the median over designs and bands the full design-to-design range.

5.5. SCEPTRE Secondary Problem

Unlike the diffusion cases above, SCEPTRE is a real, export-controlled coupled-physics code. The QoI is the net SGEMP direct-drive charge $q_{\text{net}}(\mu)$, a scalar, so no POD stage is needed. SCEPTRE does produce genuine adjoint sources for this case set, but they are deliberately not yet incorporated into GE-GPR; that integration is targeted for the December review. Every result here therefore uses standard GPR only, matching Sec. 2's value-only baseline exactly and establishing the baseline the GE-GPR extension will be measured against.

On a 1000-case void-thickness sweep (monoenergetic source) with an 80/20 held-out split, the relative error $\varepsilon_q = \|\hat{q}_{\text{net}} - q_{\text{net}}\|_2 / \|q_{\text{net}}\|_2$ falls from 1.65×10^{-3} at M=10 to 1.25×10^{-3} by M=100, then flattens at $\approx 1.15 \times 10^{-3}$ from $M \approx 200$ onward. Additional data yields essentially no further improvement past that point, consistent with q_{net} being almost linear in void thickness over the sampled range and with the floor being set by the converged white-noise level. This is the first genuine held-out accuracy result for a SCEPTRE-based QoI in this line of work, and an almost-linear 1D map is exactly the regime where gradients helped least above, so a large GE-GPR gain here should not be assumed.

The same baseline on the 1000-case void/dielectric/shield grid (10^3 factorial, all three thicknesses active,

800/200 split) behaves differently: ε_q starts at 3.32×10^{-3} at M=10, roughly double the single-parameter case, and is still declining at M=800 (1.53×10^{-3} at M=200 to 1.38×10^{-3} at M=800) where the single-parameter case has plateaued since $M \approx 200$. Both the slower convergence and the higher error at every matched M mark this as a genuinely harder map, which makes it, not the near-linear sweep, the more promising target for the GE-GPR extension.

Table II. Held-out relative error ε_q (all values $\times 10^{-3}$) for the two SCEPTRE SGEMP datasets, standard GPR only, at matched training-set sizes. $M_{\max}$ is the largest budget each split allows (799 and 800 respectively).

Dataset	M=10	50	100	200	400	$M_{\max}$
Void thickness only (1000 cases, p=1)	1.65	1.41	1.25	1.15	1.17	1.15
Void/dielectric/shield (1000 cases, p=3)	3.32	1.62	1.53	1.53	1.47	1.38

6. DISCUSSION

Read against the projection floor rather than against each other, the results fall into three regimes. Where the coefficient map is hard to learn from values alone and both surrogates sit well above the floor, gradients cut error substantially and the gap persists (q at 95.4%, the combined case at 76.8%). Where the map is easy but the design not yet dense enough, gradients yield only faster convergence, since both surrogates end at the same floor (D, and the two-parameter geometry case). Where a modest design already saturates the basis, as in the single-parameter geometry rows, nothing is left for any surrogate to gain and the 0% entries are not evidence against the method.

This is the principal caveat on the comparison. Because three of the five one-at-a-time cases are floor-limited, "the relative benefit grows with p_{act}" is not something these results establish: p_{act} and headroom-above-floor move together across the cases studied, and the present design cannot separate them. Holding the projection floor fixed while varying p_{act} is the obvious next experiment. What the combined case does establish is that the adjoint route stays economical as p_{act} grows, since gradient cost is tied to R rather than p, and that the resulting surrogate is roughly four times more accurate at a matched budget. Two further limitations bound that claim. The augmented kernel grows as $M(1 + p_{\text{act}})$ and its factorization cubically in that, which is why the combined case runs at $M = 80$; sparse or weighted variants developed for exactly this problem [4, 5] are the natural remedy. And these are fixed-basis results, so the gains reported are gains in the regression stage, not in end-to-end ROM cost.

The SCEPTRE baselines (Table II) echo the same trend in a real code: the near-linear single-parameter sweep saturates quickly, as D and the single-parameter geometry cases did, while the 3-parameter grid is measurably harder and still improving at M=800, the same qualitative shift the combined case showed going from p_{act}=1 to 5.

7. CONCLUSIONS AND FUTURE WORK

Gradient-enhanced Gaussian process regression is established prior art. The contribution of this paper is the adjoint route to the gradients it needs, in a form suited to nuclear-engineering solvers: freeze the POD basis, treat each retained coefficient as a scalar QoI linear in the state, and obtain its derivatives with respect to all active parameters from one adjoint solve per mode, Eq. (4). This provides two properties the surrounding GEK literature does not combine. Gradient cost scales with the retained rank R, typically one or two, rather than with the parameter count p, so adding parameters adds derivative observations without adding solves.

And the derivative of the snapshot field, $\partial y/\partial \mu$, is never required, which is what makes the recipe portable to a production transport code that exposes adjoint solves but not derivative fields.

Verified against exact adjoint gradients on two 1D diffusion test cases, the resulting surrogate cut median field-reconstruction error by 76.8% relative to standard GPR on a combined 5-parameter problem at matched regression budget, a separation reproduced and, if anything, strengthened (85.1% median, range 72.6–89.0%) across five independent LHS designs. On the source-term parameter the saving was roughly an order of magnitude in solves. Where a case's floor was already reached by a value-only GP, gradients bought convergence speed rather than accuracy; headroom above the floor, not parameter count, decided which regime each case fell in. Two real SCEPTRE datasets (Table II) gave this line of work its first held-out accuracy baselines outside the diffusion sandbox, using standard GPR only; the harder of the two, the 3-parameter grid, is the concrete target for the GE-GPR extension.

Future work follows from the limitations above: separating p_{act} from headroom-above-floor by holding the floor fixed while varying dimension; controlling the augmented kernel's cubic cost with sparse or weighted GEK variants [4, 5]; a multi-output GP in place of one per mode; charging basis and regression to one budget to measure end-to-end ROM cost; and, for December, wiring SCEPTRE's adjoint into GE-GPR and extending the pipeline to 3D.

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